

Title: Approximate Nearest Neighbor Search in High-Dimensional Vector Spaces at Scale

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Abstract

Approximate nearest neighbor search underpins modern retrieval systems, from semantic search engines to recommendation platforms. This paper compares graph-based, tree-based, and quantization-based indexing methods on billion-scale vector datasets, evaluating the trade-off between recall, latency, and memory footprint.

1. Introduction

As embedding dimensionality and dataset sizes grow, exact nearest neighbor search becomes computationally infeasible. Approximate methods trade a small amount of accuracy for dramatic improvements in query speed, but the optimal method depends heavily on data distribution and hardware constraints.

2. Methods Compared

We benchmark hierarchical navigable small world graphs, inverted file indexes with product quantization, and randomized KD-tree forests across three datasets ranging from ten million to one billion 768-dimensional vectors.

3. Results

Graph-based methods achieve the highest recall at a given latency budget for datasets under 100 million vectors but require substantially more memory. Quantization-based methods scale more gracefully to billion-vector datasets, sacrificing approximately 4 percent recall for a 6-fold reduction in memory usage.

4. Discussion

No single method dominates across all operating points. Systems handling diverse workloads may benefit from hybrid architectures that route queries to different index types based on latency and memory budgets.

5. Conclusion

Index selection for large-scale vector search should be treated as a joint optimization over recall, latency, and infrastructure cost rather than a fixed architectural choice.